

Ammonium Chloride Content in Aqueous Solution

Determination of the weak acid ammonium chloride in aqueous solution by titration with sodium hydroxide.

Sample	Ammonium chloride solution, 5 mL 0.1 M NH ₄ Cl solution	Preparation and Procedures 1) The titer of sodium hydroxide is determined using potassium hydrogenphthalate as a primary standard (KHP): 80 mg is dissolved into 50 ml deionized water and titrated. 2) 5 mL of 0.1 M ammonium chloride solution is added in the sample beaker. Its concentration is stored as auxiliary value H[NH₄Cl] in g/mL. 3) Dilute the sample with 50 mL deionised water and start the titration method. 4) The content is expressed as % w/w. 5) Since a standard aqueous solution of ammonium chloride was used, the result in the calculation function indicates the recovery rate of the ammonium chloride in solution.
Compound	Ammonium chloride, NH ₄ Cl, M = 53.49 g/mol, z = 1	
Chemicals	50 mL deionized water	
Titrant	Sodium hydroxide, NaOH c(NaOH) = 0.1 mol/L	
Standard	Potassium hydrogen phthalate, KHP, 80 mg	
Indication	DG111-SC	
Chemistry	NH ₄ Cl + NaOH → NH ₄ OH + NaCl	
Calculation	Content (%), expressed as w/w R1 = Q*C/m, C = M/(10*z)	
Waste disposal	Neutralization before final disposal as aqueous solution.	Remarks 1) The method parameters have been developed and optimized for the above mentioned sample. 2) If the composition of your sample is different, it may be necessary to slightly adapt the method parameters to your specific sample. 3) The method can be easily modified for automated operation. Select the appropriate sample changer in the method function "Titration stand".
Author, Version	Maria-José Schmid, MSG Anachem, July 2006	

Instruments

- T50/T70/T90 Titration Excellence, G20 Compact Titrator (with small changes).
- XS205 Balance
- Rondolino Sample Changer with diaphragm pump and PowerShower™

Accessories

- 1 x 10 mL DV1010 burette
- Titration beaker ME-101974
- Olivetti Printer JobJet 210

Results

All results

Method-ID	M401
Sample	Ammonium chloride solution (1/1)
R1 (Content)	101.193 %
Sample	Ammonium chloride solution (1/2)
R1 (Content)	101.345 %
Sample	Ammonium chloride solution (1/3)
R1 (Content)	101.466 %
Sample	Ammonium chloride solution (1/4)
R1 (Content)	101.156 %
Sample	Ammonium chloride solution (1/5)
R1 (Content)	101.305 %
Sample	Ammonium chloride solution (1/6)
R1 (Content)	101.117 %

Statistics

Method-ID	M401
R1	Content
Samples	6
Mean	101.264 %
s	0.132 %
srel	0.131 %

Titration curve

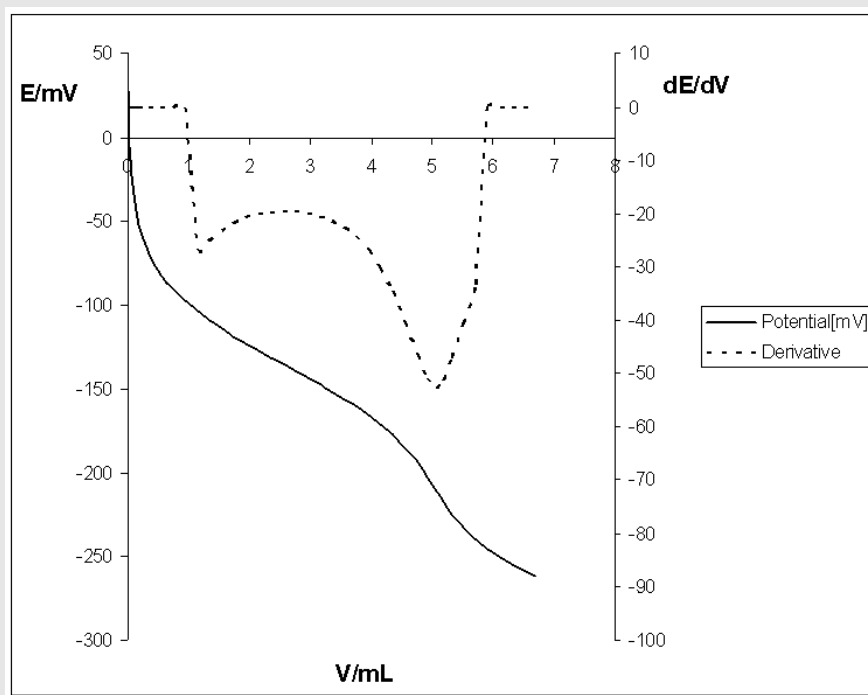


Table of measured values

Consumption / mL	Potential / mV	Derivative / dmV/dV	Time / s
0	56	NaN	0
0.02	4.4	NaN	3.6
0.04	-13	NaN	6.7
0.094	-34.3	NaN	9.7
0.184	-51.8	NaN	12.8
0.351	-69.1	NaN	15.8
0.551	-81.6	NaN	18.8
0.751	-90.6	NaN	21.9
0.951	-97.9	NaN	24.9
1.151	-104.1	-26.96	27.9
1.351	-109.5	-25.02	31
1.551	-114.4	-23.25	34
1.751	-118.9	-21.85	37.1
1.951	-123.2	-20.85	40.1
2.151	-127.4	-20.19	53.7
2.351	-131.3	-19.83	56.8
2.551	-135.3	-19.66	59.8
2.751	-139.2	-19.68	62.8
2.951	-143.2	-19.95	65.9
3.151	-147.3	-20.49	68.9
3.351	-151.4	-21.3	72
3.551	-155.8	-22.56	75
3.751	-160.5	-24.36	78.1
3.951	-165.6	-26.85	81.1
4.151	-171.3	-30.13	84.1
4.351	-177.7	-34.56	87.2
4.551	-185.1	-40.41	90.2
4.751	-193.6	-46.56	93.3
4.928	-202.4	-50.95	96.3
5.066	-210.3	-52.72	99.3
5.187	-217.1	-51.38	102.4
5.332	-224.4	-47.24	105.4
5.51	-232.4	-40.75	108.5
5.709	-239.6	-33.48	111.5
5.909	-245.5	NaN	114.6
6.109	-250.6	NaN	117.6
6.309	-254.7	NaN	120.7
6.509	-258.3	NaN	123.7
6.709	-261.3	NaN	126.7

Comments

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Method

001 Title

Type	General titration
Compatible with ID	T50 / T70 / T90 M401
Title	Ammonium chloride content
Author	METTLER TOLEDO
Date/Time	02.08.2006 15:00:00
Modified at	–
Modified by	–
Protect	No
SOP	None

002 Sample

Number of IDs	1
ID 1	Ammonium chloride solution
Entry type	Fixed volume
Volume	5 mL
Density	1.0 g/mL
Correction factor	1.0
Temperature	25.0°C

003 Titration stand (Manual stand)

Type	Manual stand
Titration stand	Manual stand 1

004 Stir

Speed	30%
Duration	10 s
Condition	No

005 Titration (EQP) [1]

Titrant

Titrant	NaOH
Concentration	0.1 mol/L

Sensor

Type	pH
Sensor	DG111-SC
Unit	mV

Temperature acquisition

Temperature acquisition	No
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Stir

Speed	30%
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Predispense

Mode	Potential
Potential	-120 mV
Wait time	10 s

Control

Control	User
Titrant addition	Dynamic
dE (set value)	8 mV
dV (min)	0.02 mL
dV (max)	0.2 mL
Meas. Val. acquisition	Equilibrium controlled
dE	1.0 mV
dt	1 s
t(min)	3 s
t(max)	30 s

Evaluation and recognition

Procedure	Standard
Threshold	20 mV/mL
Tendency	Negative
Ranges	0
Add. EQP criteria	No

Termination

At Vmax	10.0 mL
At potential	Yes
Potential	-300 mV
At slope	No
After number of recognized EQPs	Yes
Number of EQPs	1
Combined termination criteria	No

006 Calculation R1

Result	Content
Result unit	%
Formula	$R1 = Q \cdot C / (m \cdot H[NH_4Cl])$
Constant	$C = M / (10 \cdot z)$
M	M[Ammonium chloride]
z	z[Ammonium chloride]
Decimal places	3
Result limits	No
Record statistics	Yes
Extra statistical func.	No
Send to buffer	No
Condition	No

007 Record

Results	Per series
Raw results	Per series
Table of meas. values	Last titration function
Sample data	Per series
Resource data	No
E - V	Last titration function
dE/dV - V	Last titration function
log dE/dV - V	No
d ² E/dV ² - V	No
BETA - V	No
E - t	No
V - t	No
dV/dt - t	No
T - t	No
E - V & dE/dV - V	No
V - t & dV/dt - t	No
Method	No
Series data	No
Condition	No

008 End of sample

G20 Compact Titrator Method:

The G20 method includes slightly modifications:

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002 Sample

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Density	0.005349 g/mL
<i>(Since the G20 has fixed calculations and no auxiliary values, the density parameter is used to indicate the sample concentration of 0.1 M NH₄Cl)</i>	

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005 Titration (EQP) [1]

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Sensor

Type	pH
Sensor	DG115-SC
Unit	mV

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006 Calculation R1

Result type	Predefined
Calculation type	Direct titration
Formula	$R1 = Q \cdot C / (m \cdot d)$
Selected EQP	1
Result	Content
Result unit	%
Constant C	$M / (10 \cdot z)$
M	M[Ammonium chloride]
z	z[Ammonium chloride]
Decimal places	3
Result limits	No
Record statistics	Yes